

# altairsim — Quick Reference

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## Getting out, and back in

| Key               | Does                                                                        |
|-------------------|-----------------------------------------------------------------------------|
| <code>^E</code>   | <b>STOP</b> — stop the machine and take the keyboard back. Nothing is lost. |
| <code>RUN</code>  | Resume, at the exact instruction it stopped on.                             |
| <code>QUIT</code> | Leave. (There is no <code>EXIT</code> .)                                    |

## Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values ( `SET mem0 fill=` then `Tab` ); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.altairsim_history`.

| Key                                                                                            | Does                                            |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <code>Ctrl-A</code> / <code>Ctrl-E</code> (or <code>Home</code> / <code>End</code> )           | start / end of line                             |
| <code>Alt-B</code> / <code>Alt-F</code> (or <code>Ctrl-Left</code> / <code>Ctrl-Right</code> ) | back / forward one word                         |
| <code>Ctrl-W</code> / <code>Ctrl-K</code> / <code>Ctrl-U</code>                                | erase word behind / to end of line / whole line |
| <code>Backspace</code> / <code>Delete</code>                                                   | erase before / under the cursor                 |

## Command line

```
altairsim [machine] [options]
```

|                   |                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| machine           | a built-in name, or a config file (has a '/' or ends .toml).<br>Omitted: ./altairsim.toml if there is one, else `default`.                                                                                   |
| -m, --machine <n> | ALWAYS a built-in name -- never a file.                                                                                                                                                                      |
| -f, --file <path> | ALWAYS a file -- never a built-in name.                                                                                                                                                                      |
| -n, --none        | empty backplane: no boards, no memory, nothing.                                                                                                                                                              |
| -l, --list        | list the built-in machines and exit.                                                                                                                                                                         |
| -s, --script <f>  | run a command script, then exit with its status. Paths in<br>it are relative to the script's folder.                                                                                                         |
| -x, --exec <cmd>  | run one monitor command (repeatable), then exit.                                                                                                                                                             |
| -i, --interactive | after --script/--exec, stay in the monitor.                                                                                                                                                                  |
| --mcp             | MCP server on stdio.                                                                                                                                                                                         |
| --mirror <sink>   | with --mcp: mirror the console so a person can watch and<br>take over. socket:PORT (telnet in), or on macOS and Linux<br>pty or pty:LINK (open the link with a terminal program).<br>Add ?ro for watch-only. |
| --log <file>      | with --mcp: write what the guest prints to a text file, as<br>it prints it. An ordinary file; starts empty each run.                                                                                         |
| -v, --version     | print the version and exit.                                                                                                                                                                                  |
| -h, --help        | print this help and exit.                                                                                                                                                                                    |

## Monitor commands

Type the part before the bracket.

| Command      | Does                                                              | Usage                                                                                                 |
|--------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| B0[ARDS]     | List, add, or remove boards on the backplane.                     | BOARDS [LIST] ADD <type> <id> [k=v...] REMOVE <id>                                                    |
| B[REAK]      | Set a breakpoint on an address, memory/I/O access, or tape stop.  | BREAK [<addr>   MEM R W <addr>   IO R W <port>   TAPE STOP] [IF <expr>   LOADS <expr>] [TRACE ON OFF] |
| COM[PARE]    | Compare a range of memory against another address.                | COMPARE <range> <addr>                                                                                |
| C[ONFIG]     | Load or save the whole machine as a TOML file.                    | CONFIG LOAD <f.toml>   CONFIG SAVE <f.toml>                                                           |
| CONN[ECT]    | Attach a serial unit to an endpoint (console, socket, file, ...). | CONNECT <id>:<u> <endpoint>                                                                           |
| CONS[OLE]    | Show or set the host console's properties.                        | CONSOLE [<k>=<v>...]                                                                                  |
| DE[POSIT]    | Write bytes into memory at an address.                            | DEPOSIT <addr> <bytes...>                                                                             |
| DI[SASM]     | Disassemble memory into instructions.                             | DISASM [<addr> <range>] [n] [CPU=8080]                                                                |
| DISC[ONNECT] | Unplug the endpoint from a serial unit.                           | DISCONNECT <id>:<u>                                                                                   |
| D0           | Run a file of monitor commands, one per line, as if typed.        | D0 <file>                                                                                             |
| D[UMP]       | Show memory as hex and ASCII.                                     | DUMP [<addr> <range>] [WIDTH=16]                                                                      |
| E[DIT]       | Enter bytes into memory interactively from an address.            | EDIT <addr> [ROM]                                                                                     |
| EX[AMINE]    | Point the front panel at an address (and show that byte).         | EXAMINE [<addr>]                                                                                      |
| F[ILL]       | Fill a range of memory with a byte.                               | FILL <range> <byte>                                                                                   |
| HE[LP]       | Show help for a command.                                          | HELP [<command>]                                                                                      |
| H[ISTORY]    | Replay the recent instruction (or bus-cycle) history.             | HISTORY [BUS CPU] [n]                                                                                 |
| I[N]         | Read a byte from an I/O port.                                     | IN <port>                                                                                             |

|           |                                                                                     |                                                                   |
|-----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| L[OAD]    | Load a file into memory (binary, Intel hex or S-record).                            | LOAD <file> [AT <addr>] [FORMAT=BIN HEX SREC] [ROM]               |
| MA[CHINE] | Load a built-in machine by name (MACHINE none empties the backplane).               | MACHINE <name>   MACHINE none                                     |
| M[OUNT]   | Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it. | MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...] |
| MOV[E]    | Copy a range of memory to another address.                                          | MOVE <range> <dest> [ROM]                                         |
| N[EXT]    | Step one instruction, running any CALL/RST to completion.                           | NEXT                                                              |
| NO[BREAK] | Remove a breakpoint, or all of them.                                                | NOBREAK [id]                                                      |
| NOP[ASTE] | Stop a PASTE that is not finished.                                                  | NOPASTE                                                           |
| O[UT]     | Write a byte to an I/O port.                                                        | OUT <port> <byte>                                                 |

|           |                                                                       |                                                          |
|-----------|-----------------------------------------------------------------------|----------------------------------------------------------|
| PA[STE]   | Send a host file of any size to the guest's keyboard, as if pasted.   | PASTE <file>                                             |
| P[OWER]   | Power-cycle the machine -- the only thing that clears RAM.            | POWER                                                    |
| Q[UIT]    | Leave the simulator.                                                  | QUIT                                                     |
| REGI[ON]  | Add a RAM or ROM region to a memory board.                            | REGION ADD <id> type=ram rom at=<addr><br>[size= mount=] |
| RE[GS]    | Show the CPU registers (SET REG changes one).                         | REGS   SET REG <r>=<v>                                   |
| RES[ET]   | Reset the machine, keeping RAM (RESET CPU resets just the processor). | RESET [CPU]                                              |
| REST[ORE] | Load machine state back from a snapshot.                              | RESTORE <file>                                           |
| R[UN]     | Start or resume the machine, optionally at an address.                | RUN [addr]                                               |

|            |                                                                                          |                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| SA[VE]     | Write a range of memory out to a file.                                                   | SAVE <file> <range> [FORMAT=BIN HEX OCTAL PRN]                                                                                                         |
| SEA[RCH]   | Find bytes or a string in a range of memory.                                             | SEARCH <range> <bytes...> "str"                                                                                                                        |
| SE[T]      | Change a property of a board, the console, display, a register, the bus, or the machine. | SET <id>[:<u>] CONSOLE DISPLAY TERMINAL MACHINE REG BUS <k>=<v>                                                                                        |
| SH[OW]     | Display the state of a board, the bus, or the machine.                                   | SHOW <id> BOARDS BOARD <type> [UNITS] MACHINES MACHINE [<name>] BUS<br>[MAP IO IRQ CONTENTION] ROMS MOUNTS PATHS CONSOLE DISPLAY SYMBOLS CLOCK VERSION |
| SN[APSHOT] | Save the whole machine state to a file.                                                  | SNAPSHOT <file>                                                                                                                                        |
| STA[RTUP]  | Edit the machine's boot list (the commands CONFIG SAVE saves as startup = [...]).        | STARTUP [ADD <command>   REMOVE <n>   CLEAR]                                                                                                           |
| S[TEP]     | Run one instruction (or n), showing the registers after each.                            | STEP [n]                                                                                                                                               |

|                        |                                                     |                                                                                   |
|------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------|
| <code>SY[MBOLS]</code> | Load or clear a symbol table for disassembly.       | <code>SYMBOLS LOAD &lt;file&gt; [REPLACE]   SYMBOLS CLEAR</code>                  |
| <code>T[RACE]</code>   | Log every bus cycle while the machine runs.         | <code>TRACE ON OFF [file]</code><br><code>[MASK=IN,OUT,IRQ,DMA,CONTENTION]</code> |
| <code>TY[PE]</code>    | Feed text to the guest as if typed at its keyboard. | <code>TYPE "text"</code>                                                          |
| <code>U[NMOUNT]</code> | Take a disk or tape out of a drive.                 | <code>UNMOUNT &lt;id&gt;:&lt;u&gt;</code>                                         |
| <code>W[H0]</code>     | Say which board answers an address or I/O port.     | <code>WHO &lt;addr&gt;   WHO IO &lt;port&gt;</code>                               |

## Boards

### CPU

| Type              | What it is                    |
|-------------------|-------------------------------|
| <code>8080</code> | MIT's 88-CPU: 8080A CPU board |
| <code>8085</code> | Generic 8085 CPU board        |
| <code>z80</code>  | Generic Z80 CPU board         |

### Memory

| Type                  | What it is                                                   |
|-----------------------|--------------------------------------------------------------|
| <code>bankmem</code>  | Bank-switched RAM (Vector, Cromemco, North Star, ExpandoRAM) |
| <code>memory</code>   | RAM/ROM board: plain, unbanked memory regions                |
| <code>v2z80rom</code> | S100Computers V2 Z80: paged monitor EEPROM                   |

## Disk

| Type        | What it is                                                  |
|-------------|-------------------------------------------------------------|
| 16fdc       | Cromemco 16FDC: floppy controller + console UART, RDOS 2.52 |
| 64fdc       | Cromemco 64FDC: floppy controller + console UART, RDOS 3.12 |
| dcdd        | MIT 88-DCDD: 8" hard-sector floppy controller               |
| dualide     | S100Computers IDE-AB: two CompactFlash sockets for CP/M 3   |
| dualsd      | S100Computers Dual SD: two microSD sockets for CP/M 3       |
| fdcplus     | FarmTek FDC+: serial drive (drive types 6, 7)               |
| hdisk       | MIT 88-HDSK Datakeeper: Pertec hard disk controller         |
| icom        | iCOM FD3712/FD3812: 8" floppy controller with boot PROM     |
| mds         | MIT 88-MDS: 5.25" minidisk controller                       |
| mdsa        | North Star MDS-A: single-density 5.25" floppy controller    |
| mdsad       | North Star MDS-A-D: double-density 5.25" floppy controller  |
| tarbell     | Tarbell #1011: single-density floppy controller             |
| tarbelldd   | Tarbell #2022: double-density floppy controller             |
| versafloppy | SD Systems VersaFloppy I/II: WD177x floppy controller       |

## Serial

| Type    | What it is                                              |
|---------|---------------------------------------------------------|
| 2sio    | MIT 88-2SIO: two 6850 serial ports                      |
| gsio    | Generic SIO: two strap-configurable serial channels     |
| io4     | SSM IO-4: two serial + two parallel ports               |
| pmmi    | PMMI MM-103: Bell 103 modem                             |
| propio  | S100Computers Console IO: Propeller console serial port |
| sbc     | SD Systems SBC-100/200: Z80 single-board computer       |
| sio     | MIT 88-SIO: one serial port (COM2502 UART)              |
| turnkey | MIT 8800b Turnkey Module: boot PROM, serial, auto-start |

## Tape

| Type | What it is                                   |
|------|----------------------------------------------|
| acr  | MTS 88-ACR: audio cassette interface         |
| uio  | MTS 88-UIO: serial port + cassette interface |

## Parallel and printer

| Type   | What it is                                         |
|--------|----------------------------------------------------|
| 4pio   | MTS 88-4PIO: up to four 6820 parallel ports        |
| c700   | MTS 88-C700: Centronics line-printer controller    |
| d7a    | Cromemco D+7A: analog + parallel I/O, joysticks    |
| lpc    | MTS 88-LPC: 88-LP line-printer controller          |
| music6 | Newtech Model 6 Music Board: 6-bit D/A and speaker |
| pio    | MTS 88-PIO: 8-bit parallel port                    |

## Video

| Type     | What it is                                                 |
|----------|------------------------------------------------------------|
| cadzilla | CADzilla: HD63484 ACRTC graphics board with a Bt453 RAMDAC |
| dazzler  | Cromemco Dazzler: color graphics                           |
| vdb8024  | SD Systems VDB-8024: 80x24 video terminal board            |
| vdm1     | Processor Technology VDM-1: 16x64 memory-mapped video      |

## Systems

| Type | What it is                                              |
|------|---------------------------------------------------------|
| sol  | Processor Technology Sol-PC I/O: serial, keyboard, tape |

## PROM programmer

| Type | What it is                          |
|------|-------------------------------------|
| pb1  | SSM PB1: 2708/2716 EPROM programmer |

## Other

| Type       | What it is                                                  |
|------------|-------------------------------------------------------------|
| fp         | Altair front panel: the SENSE switches at IN 0FFH           |
| hostbridge | Host Bridge: file transfer to the host (not a period board) |
| rtc100     | SciTronics RTC-100: battery-backed clock/calendar           |
| ss1        | CompuPro System Support 1: interrupts, timer, clock, serial |
| virtc      | MITS 88-VI/RTC: vectored interrupts + real-time clock       |

## Machines

| Machine         | What it is                                                                                                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| 8085            | A minimal 8085 machine: an 8085 CPU, 64K of RAM, and a 2SIO console.                                                 |
| acuter          | ACUTER at F000 -- CUTER on a plain Altair, with a terminal instead of a VDM.                                         |
| altmon          | An Altair with a monitor in ROM and a terminal on it.                                                                |
| amon            | AMON 3.1 in a 4K EPROM at F000 -- Martin Eberhard's full-featured Altair monitor.                                    |
| bankmem         | A bank-switched RAM machine: a Z80, a console, and a Vector Graphic 64K bankmem.                                     |
| basic4k         | The machine Altair 4K BASIC was sold to run on: an 88-SIO Teletype, a cassette in the ACR.                           |
| basic8k         | The machine Altair 8K BASIC was sold to run on: an 88-2SIO terminal, a cassette in the ACR.                          |
| cadzilla        | An HD63484 ACRTC graphics board with a Bt453 RAMDAC in an Altair -- the bench for CADzilla.                          |
| cdbl            | The default machine with the Combo Disk Boot Loader in the PROM socket.                                              |
| compupro        | A stock Altair with a CompuPro System Support 1 board for its clock/calendar.                                        |
| cuter           | CUTER 1.3 driving a Processor Technology VDM-1 -- the real Sol/CUTS monitor.                                         |
| dazzler         | A Cromemco Dazzler in an Altair -- the bench for the S-100's first color graphics card.                              |
| default         | The machine you get when you name none: 56K, and the DBL boot PROM at FF00.                                          |
| dualide         | S100Computers "IDE-AB CF" machine -- a V2 Z80 CPU board booting CP/M 3 off a CompactFlash card.                      |
| dualidesd       | S100Computers "IDE-AB CF+ESP32" machine -- both boards, CP/M 3 on CF drives A:/B: and SD drives C:/D:.               |
| dualsd          | S100Computers "Dual SD" machine -- a V2 Z80 CPU board booting CP/M 3 off a microSD card.                             |
| icom            | iCOM FD3712 8" floppy machine -- boots CP/M 2.2 off a single-density iCOM disk.                                      |
| lineprinter-lpc | The default machine with an 88-LPC line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console.  |
| lineprinter     | The default machine with an 88-C700 line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console. |
| minidisk        | The Altair Minidisk: an 88-MDS at 08 and the MDBL boot PROM. You supply the 5.25" disk.                              |
| northstar       | North Star MDS-A single-density floppy: the controller at E800 and its boot PROM. You supply the disk.               |
| northstardd     | North Star MDS-A-D double-density floppy: the controller at E800 and its boot PROM. You supply the disk.             |
| original        | The Altair as it actually left Albuquerque.                                                                          |

|           |                                                                                              |
|-----------|----------------------------------------------------------------------------------------------|
| parallel  | The default machine with two MITS parallel boards: an 88-PIO and an 88-4PIO.                 |
| ps2       | The machine MITS Programming System II ran on: basic8k's cards, but not basic8k's bootstrap. |
| ps2int    | MITS Programming System II, WITH INTERRUPTS. ps2 with A9 down and an 88-VI/RTC in it.        |
| rombasic  | MITS Extended ROM BASIC 16K -- Extended BASIC that runs directly out of ROM.                 |
| sbc200    | SD Systems SBC-200 -- a 4 MHz Z80 single-board computer running the MSMONR21 monitor.        |
| sbc200v   | The SD Systems SBC-200 with a VDB-8024 VIDEO console: SDMONV21 on an 80x24 screen.           |
| sol20     | The Processor Technology Sol-20 -- an integrated 8080 machine, running SOLOS.                |
| tarbell   | Tarbell #1011 single-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.    |
| tarbelldd | Tarbell #2022 double-density floppy machine -- boots CP/M 2.2 the moment a disk is in it.    |
| turnkey   | The MITS 8800bt -- an Altair with a Turnkey Module where the front panel used to be.         |
| vdm1      | A Processor Technology VDM-1 in an Altair, and a demo that draws on it.                      |
| z80       | A minimal Z80 machine: a z80 CPU, 64K of RAM, and a 2SIO console.                            |

## A machine file, in one look

```
[machine]
name      = "mine"
base      = "default"      # start from a machine, and say what is DIFFERENT
startup   = ["RUN FF00"]    # the operator's own keystrokes. There is no BOOT verb.

[[board]]                                # type + a NEW id      -> ADD the card
type      = "2sio"          # type + an id from the base -> REPLACE it outright
id        = "sio0"          # NO type + an id      -> MODIFY the one already there
port      = 10              # remove = true       -> PULL THE CARD OUT

[board.unit.a]                          # a unit's own settings
connect   = "console"

[[board.region]]                        # a list the card owns (memory)
type      = "ram"
at        = 0000             # hex: it is an address
size      = "56K"           # decimal: it is a size

[[board.drive]]                         # a list the card owns (disk controllers)
unit      = 0
mount     = "cpm.dsk"        # relative to THIS FILE

[console]                              # the HOST's terminal -- not a board
strip7out = true
base      = octal           # read/print the wire class in split octal (MITS style)
```

**Paths:** a path *inside* a machine file is relative to **that file**. A path you *type* is relative to **your shell**.

## Endpoints — `CONNECT <id>:<unit> <endpoint>`

| Endpoint                                  | Is                                                                                                                                          |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <code>console</code>                      | the host terminal. Exactly one unit may hold it.                                                                                            |
| <code>null</code>                         | nowhere. Writes vanish, reads never come.                                                                                                   |
| <code>loopback</code>                     | itself — what you write comes back.                                                                                                         |
| <code>scripted</code>                     | a caller in place of a human — what MCP and the tests type into.                                                                            |
| <code>socket:PORT</code>                  | <b>listens</b> , as a raw pipe — for a program at the far end. <code>?banner</code> greets each caller.                                     |
| <code>socket:HOST:PORT</code>             | <b>calls out</b> , as a raw pipe.                                                                                                           |
| <code>telnet:PORT</code>                  | <b>listens</b> , speaking Telnet — this is telnet-in for a person. Greets each caller; <code>?banner=off</code> stops it.                   |
| <code>telnet:HOST:PORT</code>             | <b>calls out</b> , taking the telnet client's part.                                                                                         |
| <code>serial:DEVICE</code>                | a real serial port on this host.                                                                                                            |
| <code>pty</code>                          | a pseudo-terminal (macOS and Linux): open <code>/tmp/altairsim0</code> with a terminal program. <code>pty:LINK</code> selects the name.     |
| <code>in:PATH</code>                      | a host file as a reader (paper tape). <code>?cps=N</code> paces it.                                                                         |
| <code>out:PATH</code>                     | a host file as a punch — 8-bit clean, never truncating.                                                                                     |
| <code>terminal</code>                     | a window the simulator draws itself (SDL builds). <code>?emulation=vt100 adm3a vt52 h19</code> , <code>?size=COLSxROWS</code> .             |
| <code>printer:QUEUE</code>                | a real print queue on this host.                                                                                                            |
| <code>&lt;endpoint&gt; FILE</code>        | a tap: append <code> FILE</code> to any endpoint to also log the line. <code>?fmt=text</code> = plain text, only what the guest prints.     |
| <code>&lt;endpoint&gt; socket:PORT</code> | a live mirror: <code>telnet</code> in to watch and take over. <code>?ro</code> = watch-only.                                                |
| <code>&lt;endpoint&gt; pty[:LINK]</code>  | the same mirror on a pseudo-terminal (macOS and Linux): open <code>/tmp/altairsim0</code> , or <code>LINK</code> , with a terminal program. |